

Con. 9941-13.

GX-12185

(3 Hours)

[Total Marks : 80]

N.B. : (1) Question no. 1 is **compulsory**.(2) Attempt any **three** questions out of remaining **four** questions.

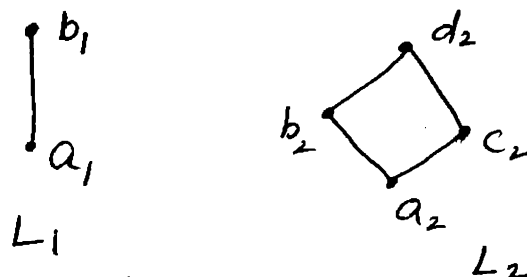
(3) Assumptions made should be clearly stated.

(4) **Figures** to the **right** indicate **full** marks.(5) Assume suitable data wherever **required** and justify it.

1. (a) Prove that in a full binary tree with n vertices, the number of pendant vertices is $(n + 1) / 2$. 4
- (b) Let G be the set of rational numbers other than 1. Let define an operation $*$ on G by $a * b = a + b - ab$ for all $a, b \in G$. Prove that $(G, *)$ is a group. 6
- (c) Find the number of integers between 1 and 1000 which are 5
 - (i) Divisible by 2, 3 or 5.
 - (ii) Divisible by 3 only but not by 2 nor by 5.
- (d) Find all solutions of the recurrence relation 5

$$a_n = 5a_{n-1} + 6a_{n-2} + 7^n$$
2. (a) Prove by mathematical induction $x^n - y^n$ is divisible by $x - y$. 4
- (b) Let m be the positive integers greater than 1. Show that the relation $R = \{(a, b) \mid a \equiv b \pmod{m}\}$, i.e. aRb if and only if m divides $a-b$, is an equivalence relation on the set of integers. 6
- (c) Let $s = \{1, 2, 3, 4\}$ and $A = S \times S$. Define the following relation :- 6

$$R \text{ on } A : (a, b) R (a', b') \text{ if and only if } a + b = a' + b'.$$
 - (i) Show that R is an equivalence relation.
 - (ii) Compute A/R .
- (d) If $f : A \rightarrow B$ be both one-to-one and onto, then prove that $f^{-1} : B \rightarrow A$ is also both one-to-one and onto. 4
3. (a) Consider an equilateral triangle whose sides are of length 3 units. If ten points are chosen lying on or inside the triangle, then show that at least two of them are no more than 1 unit apart. 5
- (b) Let L_1 and L_2 be lattices shown below :- 7

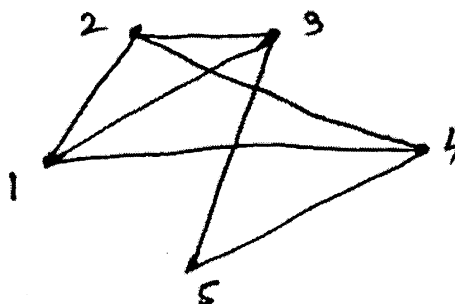
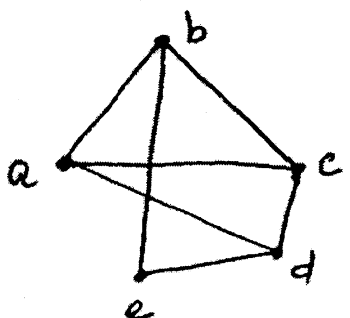
Draw the Hasse diagram of $L_1 \times L_2$ with product partial order.

- (c) Let $A = \{a, b, c\}$. Show that $(P(A), \subseteq)$ is a poset. Draw its Hasse diagram. $P(A)$ is the power set of A . 4
- (d) How many vertices are necessary to construct a graph with exactly 6 edges in which each vertex is of degree 2. 4

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4. (a) Show that if every element in a group is its own inverse, then the group must be abelian. 4
- (b) If $(G, *)$ is an abelian group, then for all $a, b \in G$, prove that by mathematical induction $(a * b)^n = a^n * b^n$. 5
- (c) If f is a homomorphism from a commutative group $(S, *)$ to another group $(T, ')$, then prove that $(T, ')$ is also commutative. 4
- (d) Consider the $(3, 5)$ group encoding function 7
 $e : B^3 \rightarrow B^5$ defined by
 $e(000) = 00000$ $e(100) = 10011$
 $e(001) = 00110$ $e(101) = 10101$
 $e(010) = 01001$ $e(110) = 11010$
 $e(011) = 01111$ $e(111) = 11100$
 Decode the following words relative to a maximum likelihood decoding function.
 (i) 11001 (ii) 01010 (iii) 00111

5. (a) Find the generating function for the following sequence 5
 1, 2, 3, 4, 5, 6,
- (b) Solve the recurrence relation $a_r = 3a_{r-1} + 2$, $r \geq 1$ with $a_0 = 1$, using generating function. 6
- (c) Show that the following graphs are isomorphic 5



- (d) Use the laws of logic to show that 4
 $[(p \Rightarrow q) \wedge \neg q] \Rightarrow \neg p$
 is a tautology
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